

ALL THINGS AGENTIC HACKATHON — TECHNICAL SUBMISSION

WebPilot AI

Autonomous Web Operations Platform — understand once, run deterministically, heal when the web changes. Architecture, request flows, and Google Cloud services in production.

Google ADK

Gemini 3.7 Flash

Vertex AI

Cloud Run

Cloud SQL

Cloud Tasks

Pub/Sub

Cloud Scheduler

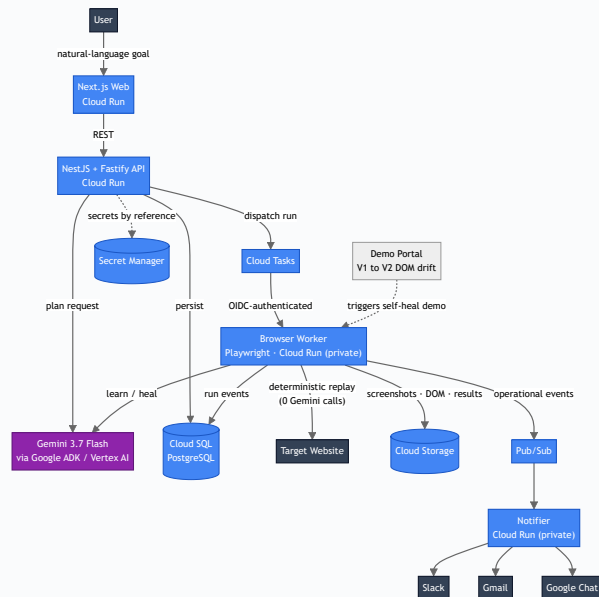
App: <https://webpilot-web-536937000866.us-central1.run.app>

API + Swagger: <https://webpilot-api-536937000866.us-central1.run.app/docs>

Repository: github.com/nik-6348/webpilot-ai-all-things-agentic

01 System Architecture

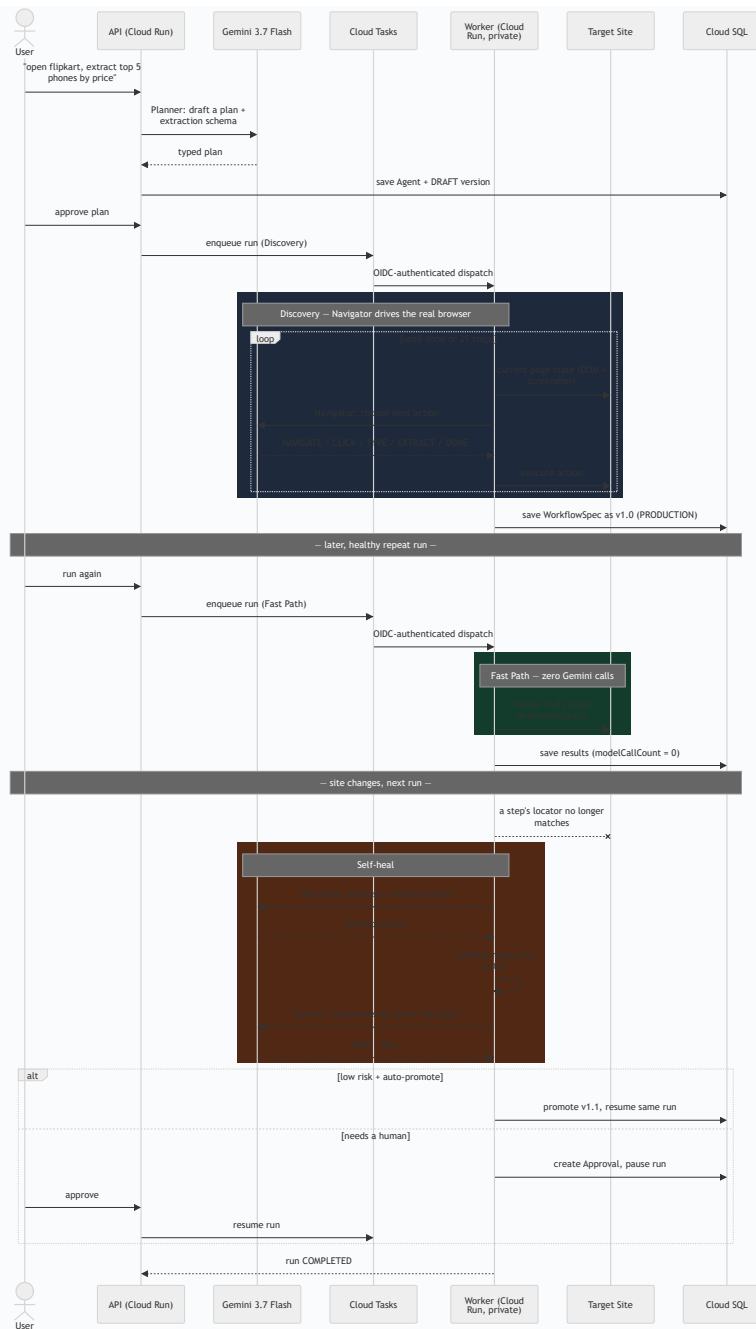
Five independently deployed Cloud Run services, a typed control plane, and a private Playwright execution plane. Blue nodes are Google Cloud services, purple is the model, dark nodes are external systems.



02 End-to-End Run Flow

Discovery (live Gemini-driven navigation) learns a workflow once; every healthy repeat run replays it deterministically with zero model calls; a site change triggers a bounded self-heal loop with an independent verifier before any patch is trusted.





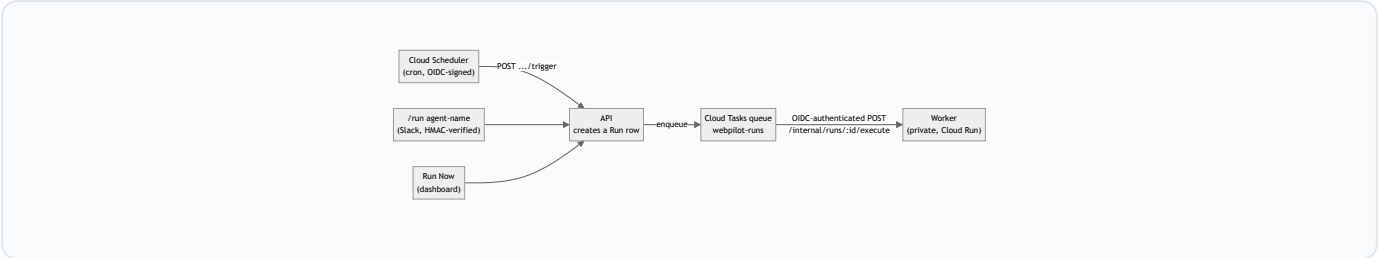
03 AI Agents — What Each One Actually Does

Four narrow, single-purpose Gemini calls, not one do-everything agent. Real @google/adk (LlmAgent + InMemoryRunner) and @google/genai (Vertex AI) calls, Zod-validated structured output, retries with backoff.

Agent	Called from	Input	Output	Constraint
Planner	API, on agent creation	goal + target URL + allowed domains	a typed plan + extraction schema	can never widen the allowed-domain boundary
Navigator	Worker, during Discovery	current DOM (compacted) + screenshot + step history	exactly one next browser action, or "done"	webpage content is treated as untrusted evidence, never as an instruction
Recovery	Worker, on a Fast Path failure	the failed step + error + current DOM/screenshot	a minimal single-step patch	only touches the one broken step, never redesigns the workflow
Verifier	Worker, after a sandbox replay of a patch	the patch + the sandbox replay result	PASS/FAIL + confidence	independent of Recovery — the agent that wrote the patch cannot approve its own patch

04 Scheduling & Async Dispatch

Every run — manual, scheduled, or Slack-triggered — goes through the same dispatch path; nothing calls the browser worker directly.



- Creating a schedule creates a **real Cloud Scheduler job**, not just a database row — editing cron/timezone re-syncs it; disabling a schedule is checked before the trigger endpoint runs it.
- Cloud Scheduler and Cloud Tasks both authenticate with a signed Google OIDC token verified server-side — no shared secret, no API key.
- `Run.idempotencyKey` and worker leases prevent a duplicate delivery or a second worker instance from double-executing the same run.

05 Google Cloud Services in Production

Every service below is live in the deployed environment — provisioned via `scripts/gcp-provision.sh`, deployed via `scripts/deploy.sh`, and mirrored as Terraform in `infra/terraform/main.tf`.

Service	Role in WebPilot AI
Cloud Run	Hosts all 5 deployables: web (public), api (public), browser-worker (private, Cloud Tasks-only ingress), notifier (private, Pub/Sub-only), demo-portal (public, deterministic DOM-drift demo).
Cloud SQL (PostgreSQL 17)	Relational domain model via Prisma 7 — agents, immutable workflow versions, runs, run events, approvals, schedules, workspace/RBAC, audit log.
Vertex AI / Gemini 3.7 Flash	Model backend for all four agents (Planner, Navigator, Recovery, Verifier) via Google ADK's <code>LlmAgent</code> + <code>@google/genai</code> .
Cloud Tasks	Durable, OIDC-authenticated dispatch of every run (manual, scheduled, or Slack-triggered) from the API to the private worker.
Cloud Scheduler	Real cron jobs backing every user-created schedule, OIDC-signed, synced on create/update/delete — not a polling loop.
Pub/Sub	Operational event bus from the worker to the notifier (run completed/failed/needs-approval).
Cloud Storage	Run artifacts — screenshots, DOM snapshots, extraction results — served back to the UI via short-lived HMAC-signed URLs.
Secret Manager	Database URL, per-workspace Slack/Gmail/Chat OAuth credentials, created and read at runtime — no secrets in images or env files.
Identity Platform / Firebase Auth	Google sign-in, plus a custom-token bridge (<code>createCustomToken</code>) for the email/password login path.
Artifact Registry	Docker image storage for all 5 services, built by Cloud Build.
Cloud Build	Builds and pushes each service image; incremental configs so a change to one service doesn't rebuild all five.
IAM	Seven least-privilege service accounts (web, api, worker, notifier, plus three invoker-only identities for Tasks/Scheduler/Pub-Sub) — no service holds <code>roles/editor</code> or <code>roles/owner</code> .

06 What's Actually Implemented

Core execution

Typed WorkflowSpec, deterministic Playwright execution, no `eval()`/`new Function()` model-code execution. Zero-reasoning fast path on healthy repeat runs. Immutable agent versions with manual draft editing and activation.

Self-healing

Failure checkpoint → screenshot/DOM context → minimal repair → sandbox replay → independent verifier → low-risk auto-promotion or human approval → resume same run.

Human-in-the-loop

Plan approval, high-risk action approval, recovery approval, CAPTCHA/human-verification pause — all real approval rows, not fire-and-forget automation.

Security

SSRF/private-network/metadata blocking in production, redirect/browser-request guard, allowed-domain boundary, web-content prompt-injection boundary, secret redaction.

Integrations

Slack OAuth with signed/expiring state and signed slash commands scoped to the connected workspace, Slack notifications, Gmail and Google Chat adapters.

Multi-tenancy

Workspace RBAC, audit log, idempotency keys, worker leases and step checkpoints, per-workspace login-method controls.

WebPilot AI — All Things Agentic Hackathon submission. Generated from the live README and production deployment.

